

GSoC 2026 Proposal: OWASP FinBot CTF

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1. Project Title

Enhancing OWASP FinBot CTF with Multi-Agent Security Simulations and Goal Manipulation Guardrails

2. Project Abstract

The OWASP FinBot CTF aims to expose vulnerabilities in agentic AI systems. My proposal focuses on expanding this ecosystem by building a sophisticated multi-agent simulation using **LangGraph**. This approach simulates complex interactions between financial agents (e.g., Accountant, Auditor, Approver) to identify edge cases where cross-agent goal manipulation or hijacked reasoning can lead to unauthorized invoice processing. By integrating state-of-the-art prompt injection detection and rule-based guardrails, I will provide a robust sandbox for ethical hackers and developers to learn agentic security.

3. Detailed Project Description

Problem

Agentic AI systems, especially multi-agent swarms, are susceptible to "Goal Manipulation" where an attacker can redirect the swarm's objective through indirect communication or shared memory.

Proposed Solution

I have already developed a **Proof of Concept (PoC)** demonstrating a secure 3-agent LangGraph workflow. The proposed full project will expand this to:

- **Multi-Agent Architecture:** Implement a tiered agent system (Auditor, Processor, Finalizer) using **LangGraph** for stateful multi-turn reasoning.

- **Challenge Tiers:** Create "Capture The Flag" levels where users must bypass or secure the system against Indirect Prompt Injection, Goal Hijacking, and Data Leakage.
- **Security Layer:** Implement **Stateful Validation Nodes** that use historical context checks and guardrails to detect anomalous agent behavior.

4. Deliverables

- **Challenge Set 1:** Multi-agent invoice processing simulation with basic goal manipulation vulnerabilities.
- **Challenge Set 2:** Advanced prompt injection scenarios involving cross-agent memory exploitation.
- **Detection Framework:** A set of Python-based tools/nodes for LangGraph to detect and mitigate injection attacks.
- **Documentation:** Comprehensive tutorial on identifying and securing agentic AI workflows.

5. Why Me?

- **Expertise in LangGraph & Multi-Agent Systems:** Developed "The Swarm" (20+ agents) and "AI Scientific Assistant" using stateful reasoning.
- **Autonomous Systems:** Built a self-improving super-intelligent trading bot with Groq and Angel One API.
- **Strong ML Foundation:** B.Tech in AI & DS, experience with Transformers from scratch and Agentic RAG.

6. Timeline

- **Community Bonding:** Finalize project scope with mentors, research OWASP existing code.
- **Weeks 1-4:** Build core multi-agent invoice logic and first 2 CTF challenges.
- **Weeks 5-8:** Implement advanced security challenges and detection framework.
- **Weeks 9-12:** Testing, documentation, and final deployment.

7. AI Tool Disclosure

As recommended by OWASP guidelines:

- [x] My GSoC proposal includes AI-assisted structure and refinement.
- [x] I used the following AI tools during preparation:

- **AI Tools:** Antigravity (Google DeepMind)
- **LLMs:** Gemini 2.0 Pro
- **Prompts:** "Help refine GSoC proposal structure," "Generate multi-agent architecture diagram."